

The Carbon Footprint of Hospital Services and Care Pathways: A State-of-the-Science Review

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BACKGROUND: Climate change is the 21st century's biggest global health threat, endangering health care systems worldwide. Health care systems, and hospital care in particular, are also major contributors to greenhouse gas emissions.

OBJECTIVES: This study used a systematic search and screening process to review the carbon footprint of hospital services and care pathways, exploring key contributing factors and outlining the rationale for chosen services and care pathways in the studies.

METHODS: This state-of-the-science review searched the MEDLINE (Ovid), Embase (Ovid), CINAHL (EBSCOhost), GreenFILE (EBSCOhost), Web of Science, Scopus, and the HealthcareLCA databases for literature published between 1 January 2000 and 1 January 2024. Gray literature was considered up to 1 January 2024. Inclusion criteria comprised original research reporting on the carbon footprint of hospital services or care pathways. Quality of evidence was assessed according to the guidelines for critical review of product life cycle assessment (LCA). PROSPERO registration number: CRD42023398527.

RESULTS: Of 5,415 records, 76 studies were included, encompassing 151 hospital services and care pathways across multiple medical specialties. Reported carbon footprints varied widely, from 0.01 kg carbon dioxide (CO₂) equivalents (kgCO₂e) for an hour of intravenously administered anesthesia to 10,200 kgCO₂e for a year of hemodialysis treatment. Travel, facilities, and consumables were key contributors to carbon footprints, whereas waste disposal had a smaller contribution. Relative importance of carbon hotspots differed per service, pathway, medical specialty, and setting. Studies employed diverse methodologies, including different LCA techniques, functional units, and system boundaries. A quarter of the studies lacked sufficient quality.

DISCUSSION: Hospital services and care pathways have a large climate impact. Quantifying the carbon footprint and identifying hotspots enables targeted and prioritized mitigation efforts. Even for similar services, the carbon footprint varies considerably between settings, underscoring the necessity of localized studies. The emerging field of health care sustainability research faces substantial methodological heterogeneity, compromising the validity and reproducibility of study results. This review informs future carbon footprint studies by highlighting understudied areas in hospital care and providing guidance for selecting specific services and pathways. <https://doi.org/10.1289/EHP14754>

Introduction

Climate change is threatening the health of billions of people alive today.¹ Climate change causes direct health impacts as a result of heatwaves and other extreme weather events and indirectly affects health through its impacts on the physical, natural, and social systems on which health depends.² Health care systems are responsible for an estimated 4.4% of global greenhouse gas (GHG) emissions, with national shares of up to 10% in high-income countries.^{3–6} Net-zero or low-carbon targets for health care systems have been set in 84 countries and areas, covering low-income countries (e.g., Malawi), middle-income countries

(e.g., Peru), and high-income countries (e.g., the UK).⁷ To inform decision-making on carbon reduction strategies, it is crucial to improve the quantitative understanding of the environmental impact of health care delivery on multiple levels.⁸

Impactful publications on sustainability in health care have shown that health care's footprint at a macro level is attributable to hotspots such as energy use, mobility, chemical substances, and disposables.^{4,9,10} In clinical practice, the use of components such as disposables, pharmaceuticals, or energy is interconnected in clinical activities, particularly within hospital services and care pathways. These can be defined as the care provided to patients by health care workers within the hospital. Examining the climate impact of individual hospital services and care pathways enables the identification of environmental hotspots and comparative advantages of clinical alternatives.⁸ Evaluating health care's footprint on a more detailed level also puts quantification analysis in a perspective that enables all stakeholders (e.g., clinicians, industry, policymakers, patients) to be part of impact reduction.

Quantifying the environmental impact of hospital services and care pathways often relies on model estimations derived from life cycle assessment (LCA) studies. LCA is a well-established method used to estimate the environmental impact of products and processes throughout their life cycle, spanning resource extraction and production to packaging, transportation, use, and waste disposal.¹¹ LCA results are highly sensitive to methodological choices.¹² Despite various guidelines and protocols for conducting, accounting, and reporting these assessments, large methodological heterogeneity exists among LCAs in health care.^{13–15} This gives rise to critical issues such as limited comparability between studies and

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other questions in the evolving field of health care environmental footprint research. First, there is a lack of clarity regarding the underlying rationale for the selection of specific hospital services and care pathways for environmental impact analysis. Second, the overall quality of many carbon footprint assessments has not been systematically and critically appraised.¹³ This lack of systematic evaluation leaves room for debate about appropriate assessment and interpretation of the resulting findings and conclusions. Third, there is limited understanding of the similarities in carbon footprints and hotspots across different hospital services, care pathways, medical specialties, and settings.

In this state-of-the-science review, we therefore aimed to answer three research questions: *a*) Which hospital services and care pathways have had their carbon footprint studied, and what rationale guided their selection? *b*) What methods have been employed to quantify the carbon footprint of hospital services and care pathways? *c*) What are the key factors contributing to the carbon footprint of hospital services and care pathways? This study focuses on one important environmental impact in particular (i.e., climate change) because the majority of existing studies on the environmental impact of health care primarily address climate impact, making the carbon footprint a well-documented and significant area for investigation¹³ while recognizing that the impact of health care on our environment is broader. Ultimately, we believe that answering these questions helps to discern potential commonalities and patterns among the carbon footprint and hotspots of hospital services, care pathways, medical specialties, and settings. We believe this will help enable the development of comprehensive strategies to reduce carbon emissions in hospital care delivery as we progress toward net-zero and low-carbon health care systems.

Methods

This review was conducted and reported according to Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines and the Standardized Technique for Assessing and Reporting Reviews of LCA (STARR-LCA) checklist (Table S1).^{16,17} This checklist is largely based on the PRISMA guideline but adapted for LCA studies, which often have large variability in methodology and reporting. The review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO; ID CRD42023398527).

Search Strategy

The following electronic databases were searched up to 1 January 2024: MEDLINE (Ovid), Embase (Ovid), Cumulative Index to Nursing and Allied Health Literature (CINAHL; EBSCOhost), GreenFILE (EBSCOhost), Web of Science, and Scopus. The search strategy was developed based on relevant keywords identified from five known relevant articles (see the section “Full search strategy” in the Supplemental Material).^{18–22} The strategy was developed by one author (L.H.J.A.K.) and peer-reviewed by two authors (D.S.K., N.H.S.W.) and consisted of two parts: *a*) environmental sustainability/carbon footprint/life cycle assessment; and *b*) hospital care/clinical pathway/surgery or related terms, synonyms, and spelling variations. Furthermore, the HealthcareLCA database was searched for “services,” “procedures,” “medical interventions,” and “investigations” until 1 January 2024 for any missing relevant studies.²³

Reference lists of included studies were screened for potential additional studies. In addition, gray literature from relevant organizations, such as governmental institutions or health care providers, was sought on their websites (see the section “Full search strategy” in the Supplemental Material).

Inclusion criteria included articles that reported the carbon footprint of hospital services or hospital care pathways. Hospital services and hospital care pathways were defined as care that patients receive that is provided to them by health care workers in the hospital. The carbon footprint had to be reported on the level of an individual hospital service (e.g., for one surgery) or care pathway (e.g., a combination of multiple services or an explicit statement that it concerned a care pathway study). Studies were also included if they reported the carbon footprint for a group of patients receiving the same service or care pathway (e.g., a hemodialysis unit). Exclusion criteria were studies reporting the carbon footprint of entire hospitals or health care systems because these do not provide information at the level of care delivery. In addition, studies that only quantified the carbon footprint of an individual product used for a hospital service, such as surgical instruments or pharmaceutical ingredients, were excluded for the same reason. Studies that included only a single component of the carbon footprint, such as waste- or travel-related emissions, were excluded because these would not provide any information on the relative importance of different hotspots.

Only original empirical research was included; reviews, opinion-based reports, commentaries, and editorials were excluded. Articles in languages other than English or Dutch were excluded. Conference abstracts were excluded. Articles published before the year 2000 were excluded because the first international standards for LCA were only published between 1997 and 2000. Inclusion and exclusion criteria are presented in Table S2.

Duplicates were identified using EndNote’s duplicate identification strategy and then manually removed. Titles and abstracts of all potentially eligible studies were screened by one reviewer (L.H.J.A.K.) using Rayyan software.²⁴ A random sample of all retrieved studies (30%) was independently screened by another reviewer. Multiple reviewers were involved in this process as indicated by the number of acronyms listed (E.S.C., B.K., D.S.K., W.J.K.H., N.H.S.W.). Any disagreements were resolved through a consensus meeting with all reviewers. Considering that there was a high level of agreement (<2% of all double-screened articles resulted in conflicting decisions), one author (L.H.J.A.K.) completed the title/abstract screening of the remaining articles. Articles that were subsequently included by one reviewer were checked by a second reviewer (N.H.S.W.). Next, all obtained full-text articles were reviewed by one reviewer (L.H.J.A.K.) and divided up among the review team for independent review by another author (E.S.C., B.K., D.S.K., W.J.K.H., N.H.S.W.). Any disagreements were resolved via consensus or through consultation with a third reviewer.

Data Extraction

A data extraction form was made and tested via the trial input of data from five articles known to be relevant.^{18–22} The following information was extracted by one author (L.H.J.A.K.) and checked by another author (E.S.C., B.K., D.S.K., W.J.K.H., N.H.S.W., L.E.S.): author(s), year, type of study, journal, description of hospital service or care pathway, study or patient population, study location, goal of the study, intended use of results, target audience, reason(s) for the selected study focus, method(s) used to quantify the environmental impact(s), including the functional unit, protocol(s) followed, software and data types, as well as outcomes regarding the total carbon footprint in kilograms of CO₂ equivalents (kgCO₂e) and its contributing factors. Contributing factors were further grouped by one researcher (L.H.J.A.K.) into the following categories, which were adapted from the Care Pathways Guidance published by the Sustainable Healthcare Coalition²⁵: travel (e.g., patient or staff travel), facilities (e.g., heating, lighting, water use, construction and maintenance of building), medical equipment (e.g., magnetic resonance imaging scanner, hemodialysis machine), medical

consumables (e.g., disposable instruments, syringes, dressings, gloves, masks), pharmaceuticals (e.g., anesthetic gases, medical drugs, pharmaceutical packaging), waste disposal (e.g., waste incineration, landfill, recycling, wastewater treatment), and other (e.g., nonmedical consumables, food, office supplies; nonmedical equipment, computers).

When estimates were available on multiple levels (i.e., for both the whole hospital unit and the individual hospital service level), information was collected for the individual service or pathway level only. When estimates were not presented at the individual level in the respective study, they were divided by the number of patients studied to obtain an estimate per patient or treatment. Missing data was requested from study authors or extracted from graphs using imaging software (ImageJ; version 1.49v).²⁶

Quality Assessment

A *pro forma* quality assessment tool by Drew et al. was used, based on Weidema's guidelines for critical review of LCAs.^{15,27} The review team consisted of an LCA expert (L.E.S.) and medical experts (L.H.J.A.K., E.S.C., J.M.K.). Three pairs of reviewers independently assessed a total of 16 quality indicators on internal and external validity, including representativeness of data and the contextualization of results; consistency of methodological choices; reporting of results and conclusions; transparency about methodological choices, assumptions, and data sources; risk of bias; and completeness in terms of system boundaries. Any disagreements in quality scores were resolved via consensus. A maximum of 35 points could be allocated among the four phases of LCA, including the goal and scope definition (13 points), inventory analysis (7 points), impact assessment (6 points), and interpretation of results (9 points). An overall quality score was calculated as a percentage of the total points allocated. A description of how the assessment tool was operationalized by our research team can be found in Table 1.

Data Synthesis

The included studies were systematically categorized and presented in a descriptive manner, organized by medical specialty and hospital service or care pathway. Some studies could be categorized into multiple medical specialties, in which case the study was categorized to the medical specialty that the study authors emphasized. For each category of contributing factors, the relative contribution per service or pathway was calculated. In some studies, the relative contributions of contributing factors were not reported separately. If it was not possible to group a factor into one category, the total contribution of this factor was attributed to each applicable category (i.e., double counting).

Results

Search Results

A total of 9,799 records were identified from six databases (Figure 1). After removing duplicates, 5,415 records were screened, of which 5,317 were deemed irrelevant based on title and abstract. Ninety-eight articles were assessed for eligibility based on full-text review, after which 62 articles were included in the review. The search identified 14 additional relevant studies from the HealthcareLCA database, gray literature, and citation searching. In total, 76 studies were included, the majority of them published in the last 3 y ($n = 49$).

Hospital Services and Care Pathways Studied

Carbon footprints have been evaluated for a total of 151 different hospital services and care pathways and variants, as presented in

Table 2. Study topics involved surgical care, including anesthesiology ($n = 25$); interventions from different medical specialties, such as obstetrics and gynecology ($n = 9$) and urology ($n = 5$); renal care ($n = 16$); diagnostic services ($n = 28$), among which a majority focused on medical imaging ($n = 10$) and blood and urine testing ($n = 11$); ophthalmological care, including cataract surgery ($n = 11$); cardiac care ($n = 6$); emergency care ($n = 8$); outpatient care ($n = 19$); and psychiatric treatments ($n = 3$). The majority of studies focused on distinct hospital services ($n = 59$), whereas 17 studies focused on entire hospital care pathways, such as the treatment of patients with acute decompensated heart failure or hemodialysis treatment during a whole year.^{18,19,22,32,34,46,51,60,63,67,69,74–77,80,98} (Table 2). Care pathway studies generally included multiple elements of care in the functional units of their analyses, such as diagnostics and follow-up care, and evaluated care during longer periods of time, such as the duration of a hospitalization or treatment during one or multiple years.^{18,19,22,32,34,46,51,60,63,67,69,74–77,80,98}

Comparing the Carbon Footprint between Services, Medical Specialties, and Settings

The reported carbon footprints ranged between 0.01 kg carbon dioxide (CO₂) equivalents (kgCO₂e) for an hour of propofol anesthesia,²⁹ up to 10,200 kgCO₂e for a year of hemodialysis treatment.²⁰ Given the differences in functional units and included elements of care, it was difficult to directly compare the magnitude of the carbon footprints between different studies. However, a number of studies ($n = 20$) compared similar services or modalities within their study, employing similar units of analyses, which made comparisons between services more informative.^{21,28,29,32,37,41,43,44,53,60,62–65,67,69–71,85,89} Six studies evaluated the same service or pathway over time,^{34,76,77,84,96,97} and 16 studies looked across different settings.^{20,39,45,48,53,55,77,81,88,90–95,98} The majority of studies that compared care delivery at home vs. in hospital showed that home-based care delivery had a lower footprint (i.e., for hemodialysis^{45,48} and outpatient consultations).^{90–95} Regarding different facilities, one study showed that cataract surgery in a private hospital yielded a slightly larger footprint compared with such surgery in a public facility in the same region,⁵⁵ whereas another study demonstrated that the carbon footprint of skin cancer surgery was lower in a private facility.⁸⁸ One study showed that the carbon footprint of hemodialysis treatment varied substantially across 15 facilities in a single geographic region in the US.²⁰ Two studies conducted cross-country comparisons, showing that certain forms of surgical and intensive care unit care had the highest footprint in the US, compared with Canada, the UK, and Australia.^{39,81}

Rationale for Selected Hospital Services and Care Pathways

Several reasons were mentioned for selecting hospital services and care pathways for carbon footprint quantification. The majority of studies ($n = 36$) rationalized the selected study topic by indicating the high volume of patients or services provided each year,^{19,21,22,28,30,37,39,40,42–44,52–59,61,62,65–67,70,71,73,74,76,80,84,85,87–89} or an increasing (future) demand for care ($n = 11$).^{45,48,59,61,70,85,87,91,93,97,98} Five studies mentioned high volumes of waste.^{35,42,54,55,88} Others stated that the hospital service under study was particularly resource intensive in terms of, for example, energy use, water, or costs.^{21,30,33,35,41,45,49,50,68,72,78,81,86,88} Some suggested that it would be interesting to evaluate environmental outcomes, especially when clinical alternatives for services and pathways were available and equally (cost-)effective.^{18,21,28,41,45,62,74,92} Another frequently mentioned argument was the absence of carbon footprint assessments for the hospital service under study.^{31,33,34,40,42,44,64–66,68,71,73,80,83,84,86,87} One study mentioned the

Table 1. Quality assessment tool, based on Drew et al.,¹⁵ including its operationalization in this study.

Appraisal criteria	Indicator(s)	Operationalization by our research team
Phase 1: Goal and scope (13 points)		
Study goal is clearly stated, including the study's rationale (1), intended application (1), and intended audience (1)	Transparency	—
LCA method is clearly stated (1)	Transparency	If the term LCA was not explicitly used, this item scored zero points.
Functional unit is clearly defined and measurable (1), justified (1), and consistent with the study's intended application (1)	Consistency	No points were subtracted if the term "functional unit" was not explicitly used. Points were given based on a clear description of the unit of analysis. In case no intended application was mentioned (scoring item 1), consistency with the study's aim was assessed.
The system to be studied is adequately described with clearly stated system boundaries (1), life cycle stages (1), and appropriate justification of any omitted stages (1)	Transparency; bias	Points for appropriate justification of any omitted stages were not given if the study listed only excluded elements, without an explanation of why these were excluded.
The system covers production (1), use/reuse (1), and disposal (1) of materials and energy	Internal validity, completeness	The original assessment tool included: "half mark if only for energy and vice versa," which was unclear for our reviewers and left out of the assessment.
Phase 2: Inventory analysis (7 points)		
The data collection process is clearly explained, including the source(s) of foreground material weights and energy values (1); the source(s) of reference data (e.g., inventory database; 1); and what data are included (e.g., production and disposal of unit processes; 1)	Transparency, internal validity	—
Representativeness of the data is discussed (1), differences in electricity generating mix are accounted for (1), and the potential significance of exclusions or assumptions is addressed (1)	Internal validity; external validity	Point for electricity generating mix given if analyses were adjusted for local energy mix or if sensitivity to different energy mixes was assessed. Point for representativeness given only when explicitly mentioned with regard to either geographic, temporal, or technological representativeness, e.g., when prices were deflated. If geographical representativeness was only addressed in the context of the energy mix, only one point was given for "differences in electricity generating mix are accounted for." Point for addressing the potential significance of exclusions or assumptions given only if potential significance was explicitly stated, i.e., whether it potentially led to an under- or overestimation.
Allocation procedures, where necessary, are described and appropriately justified (1; mark given if no allocation was used)	Transparency; bias	This item was given a score of 1 if no substantial allocation was deemed necessary in the study.
Phase 3: Impact assessment (6 points)		
Impact categories (1), characterization method (1), and software used (1) are documented transparently	Transparency	Given that all articles mentioned the term "carbon footprint" (because this was included in the literature search), 1 point was always given for "impact categories."
Results are clearly reported in the context of the functional unit (1) (0.5 if graphically, 0 if only normalized results were reported)	Consistency; transparency	If no functional unit was described in phase 1, this item was judged based on the way results were presented in general.
A contribution analysis is performed and clearly reported (1), and hotspots are identified (1)	—	A point was given for contribution analysis if the results were summed up and presented as a total footprint.
Phase 4: Interpretation (9 points)		
Conclusions are consistent with the goal and scope (1) and supported by the impact assessment results (1)	Internal validity; consistency	If no goal and scope were described earlier, this item was judged based on the clearness of the provided conclusion(s) in general.
Results are contextualized through the use of sensitivity analysis (1) and uncertainty analysis (1)	Internal validity	If the study did not explicitly mention "sensitivity" or "uncertainty analysis," but presented ranges or standard deviations: 1 point was given for "uncertainty analysis."
Limitations are adequately discussed (1), and the potential impact of omissions or assumptions on the study's outcomes are described (1)	Bias	Only points given when the potential impact of the omission on the study's outcomes were explicitly mentioned, i.e., whether the omission likely led to an under- or overestimation.
The assessment has been critically appraised (i.e., peer review if journal article or independent, external critical review if report/thesis; 1)	Bias	No point was given in the case of a letter to the editor or a commentary because these are generally not peer-reviewed. However, if an included letter is peer-reviewed, 1 point will be given.
Source(s) of funding and any potential conflict(s) of interest are disclosed (1) and are unlikely to be a source of bias (1)	Bias	No point was given for the first item if only conflict(s) of interest were disclosed but no source(s) of funding were reported.

Note: —, not applicable; LCA, life cycle assessment.

duration of illness,⁶⁷ 2 mentioned the availability of data,^{18,92} and 10 referred to previous research that either suggested a potentially large footprint of certain types of care^{20,22,29,77,95–97} or a need for further research.^{28,75,77,91,95–97} Finally, 13 studies did not clarify their reasons for selecting certain hospital services or care pathways.^{32,36,38,46,47,51,60,63,69,79,82,90,94}

Applied Methods

Various methods were used for carbon footprint quantification, which were differently described by included studies. Table 3 lists the methods as described by the included articles. Most studies ($n = 32$) used an LCA technique, for which some used the

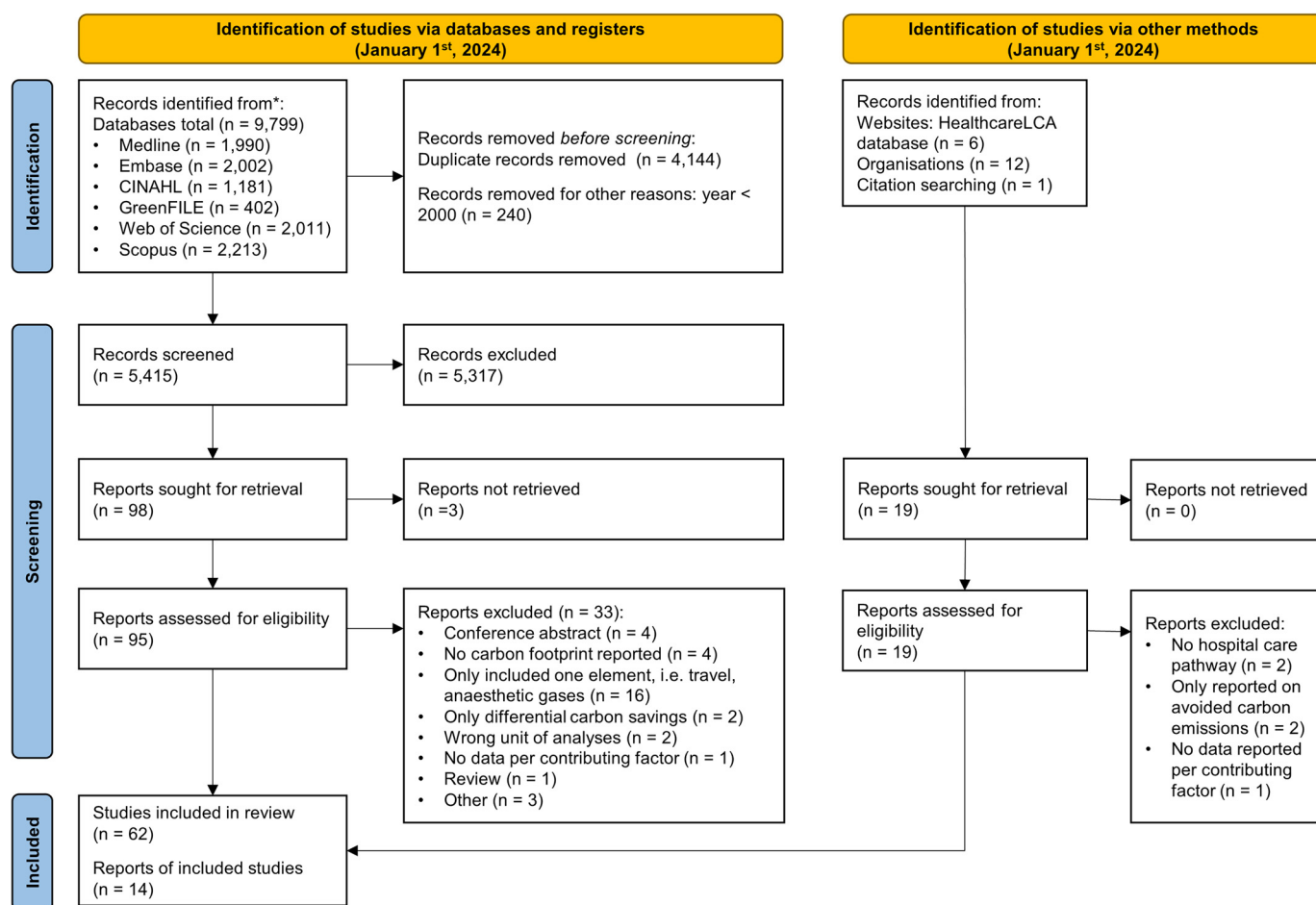


Figure 1. PRISMA Flow diagram. Note: CINAHL, Cumulative Index to Nursing and Allied Health Literature; PRISMA, Preferred Reporting Items for Systematic reviews and Meta-Analyses.

term life cycle analysis^{54,59}; both abbreviated as “LCA.” EIO-LCA was used as an abbreviation for both economic input-output LCA and environment input-output LCA,^{52,87} and environmentally extended input-output (EEIO) was used for environmentally extended input-output LCA.^{22,72} Several studies used (multi-) component analysis (n = 5)^{19,46,50,79,98} or Bilan Carbone methodology/software (n = 4).^{42,49,52,86} Although terminology varied between studies, methodological steps and approaches seemed quite similar, where most studies used process activity data (n = 40) or combined process and financial activity data in a hybrid analysis (n = 26). Only two studies used solely financial activity data, and they were both care pathway studies.^{18,22}

Contributing Factors

Table 4 shows what type of factors were included per hospital service or care pathway and their relative contributions, presented by medical specialty. More detailed descriptions of included factors can be found in Tables S3–S21. Figure 2 graphically summarizes the results for six different medical specialties. These figures can be interpreted as the share of services and pathways, as currently studied within a medical specialty, that included certain contributing factors and how often they ranked in the top 2 of the largest contributions. A large variation between medical specialties existed in the type of factors that were included and the ones that prevailed as the first or second largest carbon hotspots for services studied within that specialty. In the next sections, results will be discussed per contributing factor.

Travel. Travel was included in 75 services (covered in 45 studies) of which most services included patient travel (n = 66) and staff travel (n = 44). Four services also included visitor travel.^{36,46} Within each identified study that included travel, travel in general, and patient travel in particular, had relatively large contributions to the carbon footprint of hospital services and care pathways. Patient travel ranked as the first or second largest contributor in 67% of the services that included this factor (patient travel: n = 44 of 66 services, contribution range: 0.5%–100%; staff travel: n = 19 of 44 services, contribution range: 0.1%–75%). Results differed by medical specialty. For example, within nephrology, patient travel was included for nearly every studied hospital service and pathway (Figure 2A), but in only a part of these, it ranked as the first or second largest hotspot. In contrast, for services within pathology and radiology, patient travel was never included (Figure 2C,D).

Facilities. Facilities that are needed for delivering hospital care were included in 84 services (48 studies), ranking as the first or second largest contributor in 61% of the services that included this factor (n = 51 of 84 services, contribution range: 0.4%–100%). Most studies included only energy use of the building, such as electricity used for lighting or heating, ventilation, and air-conditioning. Some studies included the carbon footprint of water use or wastewater treatment,^{19,20,22,36–38,46,48–51,57,67,82,88,90,98} which almost always had low contributions (<4%) to the carbon footprint. Only a few studies included other facility-related factors such as construction, sanitation, or sewerage treatment.^{46,50,51,98} For certain medical specialties, including nephrology and surgical care, the facilities-related footprint was especially large (Figure 2A,E).

Table 2. Hospital services and care pathways studied, by medical specialty.

Medical specialty	Hospital service or care pathway	Care pathway ^a	Diagnostics ^b	Treatment ^b	Inpatient care ^b	Outpatient visits ^b	Specifics under study ^c	Studies (n)	Carbon footprint (kgCO ₂ e)	Reference(s)
Anesthesiology	Anesthesia for total knee replacement	—	—	Y	—	—	General; General + spinal; Spinal	1	14.9–18.5	McGain et al. ²⁸
	Maintenance anesthesia for 1 h	—	—	Y	—	—	Desflurane; Isoflurane; Propofol; Sevoflurane	1	0.01–56.5	Sherman et al. ²⁹
Cardiology and cardiac surgery	Atrial fibrillation catheter ablation procedure	—	—	Y	—	—	—	1	76.9	Ditac et al. ³⁰
	Cardiac surgery	—	—	Y	—	—	—	1	124.3	Grinberg et al. ³¹
	Cardioversion care	Y	Y	Y	Y	Y	DOAC; Warfarin	1	58.2–85.5	Orton and Pierce ³²
	Cardioversion care pathway	Y	Y	Y	Y	—	—	1	263	Zhang et al. ²²
	Care pathway for acute decompensated heart failure	—	—	Y	—	—	—	1	505.1	Hubert et al. ³³
Dermatology	Elective CABG with cardiac bypass	—	—	Y	—	Y	Various melanoma stages (1a–3d), following 2015 vs. 2022 guidelines	1	157.9–1,657.7	Grover et al. ³⁴
	Melanoma follow-up pathway	Y	Y	—	—	—	High-intensity ward; low-intensity ward; in ICU; in acute care unit	3	12–161.6	Hunfeld et al. ³⁵ ; Penny et al. ³⁶ ; Prasad et al. ³⁷
Emergency medicine	Bed day	—	—	Y	Y	—	—	1	13.8	Penny et al. ³⁸
	Emergency department visit	—	Y ^d	Y	—	—	—	1	88–178	McGain et al. ³⁹
	Treatment of septic shock in ICU	—	Y	Y	Y	—	American hospital; Australian hospital	1	5.4	Elli et al. ⁴⁰
Gastroenterology	Colonoscopy	—	Y	—	—	—	With vs. without	2	0.5–6.7	Elli et al. ⁴⁰ ; Zullo et al. ⁴¹
	Esophagogastroduodenoscopy/upper endoscopy	—	Y	—	—	—	EndoFaster to reduce rate of biopsy	1	28.4	Lacroute et al. ⁴²
	Gastrointestinal endoscopy	—	Y	—	—	—	Average of different types of endoscopy, including gastroscopy and colonoscopy	1	247	Gatenby ¹⁸
Gynecology and obstetrics	Medical treatment of gastroesophageal reflux	Y	—	Y	Y	Y	—	1	37	Campion et al. ⁴³
	Cesarean section	—	—	Y	—	—	—	1	22.7–40.3	Woods et al. ⁴⁴
	Endometrial cancer staging	—	—	Y	—	—	Laparoscopy; laparotomy; robot-assisted	1	285–814	Thiel et al. ²¹
	Hysterectomy	—	—	Y	—	—	Abdominal; laparoscopic; robotic; vaginal	1	17	Campion et al. ⁴³
Nephrology	Vaginal birth	—	—	Y	—	?	Continuous, at home; continuous, in PD center; daytime, at home; daytime, in PD center	1	363.5–409.5	Chen et al. ⁴⁵
	Ambulatory PD	—	—	Y	?	—	With Prismaflex system	1	161	Connor et al. ⁴⁶
	Bed day nephrology ward	—	Y	Y	Y	—	—	1	113	Aspin ⁴⁷
	Continuous renal replacement therapy (per 72 h)	—	—	Y	—	—	—	1	58.9	Sehgal et al. ²⁰
	Hemodialysis (per treatment)	—	—	Y	—	—	Unit-based; home-based	1	207–1,404	James ⁴⁸
	Hemodialysis (per year)	—	—	Y	—	Y	At home; in center	1	5,110	Mitoui et al. ⁴⁹
	Hemodialysis (per year)	—	Y	Y	—	—	—	1	10,200	Lim et al. ⁵⁰
	Hemodialysis care pathway (per year)	Y	—	Y	—	Y	—	1	1,844	Connor et al. ⁴⁶

Table 2. (Continued.)

Medical specialty	Hospital service or care pathway	Care pathway ^a	Diagnostics ^b	Treatment ^b	Inpatient care ^b	Outpatient visits ^b	Specifics under study ^c	Studies (n)	Carbon footprint (kgCO ₂ e)	Reference(s)
Ophthalmology	Hemodialysis care pathway (per year)	Y	Y	Y	—	—	—	1	3,382	Newcastle upon Tyne Hospital ⁵¹
	Hemodialysis and PD (per year)	Y	Y	Y	—	—	—	1	7,094	Connor et al. ⁴⁶
	Outpatient appointment	—	?	?	—	Y	Manual small incision cataract surgery; phacoemulsification; phacoemulsification at public hospital; phacoemulsification in private hospital; unspecified	1	22	Connor et al. ⁴⁶
	Cataract surgery	—	—	Y	—	—	—	7	5.9–158.6	Ferrero et al. ⁵² ; Goel et al. ⁵³ ; Hong et al. ⁵⁴ ; Latta et al. ⁵⁵ ; Pascual-Prieto et al. ⁵⁶ ; Thiel et al. ⁵⁷
Orthopedics	Cataract pathway	Y	—	Y	—	Y	—	1	181.8	Morris et al. ¹⁹
	Intravitreal injection	—	—	Y	—	—	—	2	13.7–14.1	Chandra et al. ⁵⁸ ; Power et al. ⁵⁹
	Total knee replacement care pathway	Y	Y	Y	Y	Y	With Care4Today program; without Care4Today program	1	336–421	Johnson & Johnson ⁶⁰
	Total knee replacement surgery	—	—	Y	—	—	Posterior-stabilized cemented, with tibial extension stem, performed by a medial parapatellar approach	1	190.5	Delaie et al. ⁶¹
Otolaryngology	Tonsillectomy	—	—	Y	—	—	Monopolar electrocautery; cold excision without cautery; coblation	1	157.6–204.7	Meiklejohn et al. ⁶²
Pediatric medicine	Pediatric asthma care pathway (per year)	Y	?	Y	Y	?	With Smartinhaler; without Smartinhaler	1	39–89	Budgen ⁶³
Pathology	Blood testing	—	Y	—	—	—	ABG; CRP; coagulation profile; full blood examination; urea + electrolytes; chemistry; coagulation factors; CBC + differential; total protein; phlebotomy	2	0.04–0.7	McAlister et al. ⁶⁴ ; Spoyalo et al. ⁶⁵
Psychiatry	Urine testing	—	Y	—	—	—	Urinalysis	—	—	—
	COVID-19 testing	—	Y	—	—	—	—	1	0.6	Ji et al. ⁶⁶
Pulmonary medicine	Schizophrenia pathway with antipsychotic injections	Y	—	Y	Y	Y	One-monthly; three-monthly; treatment interruption	1	364.9–766.5	Debaveye et al. ⁶⁷
	Bronchoscopy	—	Y	—	—	—	—	1	1.1	Patrucco et al. ⁶⁸
	Monoclonal antibody therapy of severe eosinophilic asthma	Y	?	Y	Y	?	With benralizumab; without benralizumab	1	91–192	Budgen ⁶⁹
Radiology	CT scan	—	Y	—	—	—	—	2	6.7–9.2	Martin et al. ⁷⁰
	Chest X-ray	—	Y	—	—	—	Nonmobile; mobile	1	0.5–0.8	McAlister et al. ⁷¹
	Interventional radiology procedure	—	Y	Y	—	—	—	1	243	Chua et al. ⁷²

Table 2. (Continued.)

Medical specialty	Hospital service or care pathway	Care pathway ^a	Diagnostics ^b	Treatment ^b	Inpatient care ^b	Outpatient visits ^b	Specifics under study ^c	Studies (n)	Carbon footprint (kgCO ₂ e)	Reference(s)
Radiotherapy	MRI scan	—	Y	—	—	—	Prostate; not specified	3	17.5–22.4	Esmaili et al. ⁷³ , Leapman et al. ⁷⁴ , Martin et al. ⁷⁰ , McAlister et al. ⁷¹
	Ultrasound	—	Y	—	—	—	—	2	0.5–1.1	Martin et al. ⁷⁰ , McAlister et al. ⁷¹
	External beam radiation therapy	Y	—	Y	—	Y	Average treatment course (various disease sites); during pandemic; pre-pandemic (various treatment courses)	2	249–489	Ali and Piffoux ⁷⁵ , Cheung et al. ⁷⁶
	External beam radiation therapy	Y	Y	Y	—	—	Prostate VMAT, pre-COVID; prostate VMAT, during COVID; breast IMRT, pre-COVID; breast IMRT, during COVID	1	75–227	Chuter et al. ⁷⁷
	Proton therapy	—	—	Y	—	—	Various disease sites treated, e.g., breast cancer, prostate cancer	1	1,361–1,409	Dvorak et al. ⁷⁸
	Abdominoplasty	—	—	Y	?	—	—	1	23.7	Berner et al. ⁷⁹
Surgery	Anastomotic leak care pathway	Y	Y	Y	Y	Y	Grades A, B, and C	1	1,302.97	Bischofberger et al. ⁸⁰
	Anti-reflux surgery	Y	Y	Y	Y	Y	—	1	1,038	Gatenby ¹⁸
	Average surgery	—	—	Y	—	—	American hospital; Canadian hospital; UK hospital; hospital in low- or middle-income country	3	8.4–232	MacNeill et al. ⁸¹ , Penny et al. ⁸² , Umo et al. ⁸³
	Bilateral breast augmentation	—	—	Y	?	—	—	1	16.2	Berner et al. ⁷⁹
	Carpal tunnel release	—	—	Y	—	—	Standard vs. lean and green; endoscopic; open	2	6.6–106.5	Kodumuri et al. ⁸⁴ , Zhang et al. ⁸⁵
	Elective endovascular aortic repair	—	—	Y	—	—	—	1	108	Sénémaud et al. ⁸⁶
Urology	Minimally invasive surgery	—	—	Y	—	—	—	1	141	Power et al. ⁸⁷
	Rhinoplasty	—	—	Y	?	—	—	1	17	Berner et al. ⁷⁹
	Skin cancer surgery	—	—	Y	—	—	Clinic-based; hospital-based	1	28.5–80.8	Tan and Lim ⁸⁸
	Prostate biopsy pathway (included 3 separate services; prebiopsy prostate MRI, TRUS biopsy, and pathology analysis)	Y	Y	—	—	—	—	1	36.2–80.7	Leapman et al. ⁷⁴
	Radical prostatectomy	—	—	Y	Y	—	Laparoscopic; robot-assisted	1	47.3–59.7	Fuschi et al. ⁸⁹

Table 2. (Continued.)

Medical specialty	Hospital service or care pathway	Care pathway ^a	Diagnostics ^b	Treatment ^b	Inpatient care ^b	Outpatient visits ^b	Specifics under study ^c	Studies (n)	Carbon footprint (kgCO ₂ e)	Reference(s)
Outpatient consultations in multiple medical specialties	Face-to-face consultation	—	—	?	—	Y	Geriatric medicine clinic; psychiatric clinic; rehabilitation clinic; speech therapy clinic; several different departments, i.e., urology	5	4.8–178.5	Bartlett and Keir ⁹⁰ ; Filfilan et al. ⁹¹ ; Holmner et al. ⁹² ; Penaskovic et al. ⁹³ ; Sillico et al. ⁹⁴ ; Thiel et al. ⁹⁵
	Neuroemergent consultation	—	—	—	—	Y	With telehealth; without telehealth	1	131–437	Whetten et al. ⁹⁶
	Preoperative screening	—	Y	—	—	Y	After telehealth implementation; before telehealth implementation	1	76.4–84.5	Wang et al. ⁹⁷
	Virtual consultation	—	—	—	—	Y	By phone; by video; geriatric medicine clinic; psychiatric clinic; several different departments, i.e., urology	6	0–4.1	Bartlett and Keir ⁹⁰ ; Filfilan et al. ⁹¹ ; Holmner et al. ⁹² ; Penaskovic et al. ⁹³ ; Sillico et al. ⁹⁴ ; Thiel et al. ⁹⁵

Note: Results should not be directly compared due to differing methodologies. —, Not applicable; ABG, arterial blood gas; CABG, coronary artery bypass graft; CBC, Complete blood count; COVID, Coronavirus disease; CRP, C-reactive protein; CT, Computerized Tomography; DOAC, direct oral anticoagulant; ICU, Intensive Care Unit; IMRT, Intensity-modulated radiotherapy; MRI, magnetic resonance imaging; PD, peritoneal dialysis; TRUS, transrectal ultrasound; VMAT, Volumetric modulated arc therapy.

^aStudies that evaluated care pathways are indicated with “Y.”

^b“Y” indicates that this specified care type was included in the analysis.

^cSeveral studies included specific details of the hospital service or care pathway under study, for example, about the setting in which the service was provided or which variants were compared. In case a study specified these, the details are listed in this column.

^d“—” indicates that in some studies it was not entirely clear whether the analysis included an element of care in the functional units of their analyses.

Medical equipment. One hundred thirteen services (in 57 studies) included GHG emissions from various types of medical equipment, including reusable surgical instruments, anesthesia machines, imaging equipment, and patient air warmers. For most services, only the equipment’s energy use was included, but some also included equipment production and sterilization of reusable equipment. Medical equipment ranked as the first or second largest contributor to the carbon footprint within certain medical specialties, such as radiology (Figure 2D). However, for most other specialties and hospital services, equipment did not contribute as much. Overall, medical equipment ranked first or second in 56% of the cases ($n = 64$ of 113 services, contribution range: 0.02%–100%).

Medical consumables. Medical consumables were included in 100 services (in 53 studies). A variety of medical consumables were included, such as drapes, gowns, gloves, syringes, packaging, medical injection packs, and other single-use medical supplies. Some studies found relatively large contributions for certain consumable items, such as packaging,⁴⁵ whereas others found that packaging had only a small impact on the carbon footprint.⁹⁸ For many hospital services, consumables ranked in the top two largest contributors (69% of the services/pathways that included consumables; $n = 69$ of 100 services, contribution range: 0.3%–89%). Interestingly, consumables were included in all services studied within ophthalmology and pathology, ranking first or second in all services except for one (Figure 2B,C). Within other specialties, including surgery, consumables are less often ranked first or second (Figure 2E).

Pharmaceuticals. Pharmaceuticals were included in 50 services (in 31 studies), ranking first or second place in 62% of the services that included pharmaceuticals ($n = 31$ of 50 services, range: 0.1%–100%). Some studies included specific pharmaceuticals such as anesthetics,^{28,29,62,81} proton pump inhibitors,¹⁸ antibiotics,⁸⁰ or antipsychotic injections.⁶⁷ Others tried including pharmaceuticals based on financial activity data, but the relative contributions were so high that they were eventually not included in the footprints.^{39,58,59} In many specialties, pharmaceuticals were rarely or never included, including nephrology, urology, and outpatient consultations in different medical specialties. The studies that included pharmaceuticals found that their contributions were relatively large, especially for anesthetic gases.^{21,62,81} On the contrary, a study that modeled the footprint of psychiatric medication injections found pharmaceuticals to be responsible for only a small part of the total footprint of the year-long pathway.⁶⁷

Waste disposal. Waste disposal was included in 97 services (in 50 studies), with an average relative contribution lower than 5% (range –3% up to 77%), ranking first or second place in 38% of services that included disposal in the analyses ($n = 37$ of 97). Studies included different forms of waste, such as landfill waste, cytotoxic waste, municipal solid waste, sharps waste, and body tissue waste, as well as different forms of waste disposal, including incineration and recycling. Some studies also included transportation to the disposal site. A few studies found that disposal had a negative contribution to the carbon footprint.^{45,50,57,88} This was explained as due to recycling or the capture of methane at landfills.⁸⁸

Other. Finally, many studies included other factors such as laundry services, cleaning compounds, internet use, nonmedical equipment, such as computers, and nonmedical consumables, including food, paper, and office supplies. The relative contributions of these factors can be found in Tables S3–S21.

Quality Assessment

Methodological quality was heterogeneous, and scores varied between 20% and 97% of total possible points. In many studies, quality was found to be insufficient, with one-quarter of studies

Table 3. Methods used for carbon footprint quantification.

Method as described by study	Studies (n)	Medical specialty	Protocol(s)	Type(s) of activity data	Software	Characterization method(s)
Life cycle assessment or similar approaches						
Life cycle assessment	19	Anesthesiology; emergency medicine; nephrology; orthopedics; pathology; radiology; surgery; urology; outpatient consultations in multiple medical specialties	GHG protocol; ISO 04014; ISO 14040; ISO 14044; SCPG; not stated	Process; hybrid; Not stated	Ecodesign Studio software; JMP Pro 15; OpenLCA; Python; SimaPro; not stated	GWP100 work of Sulbaek Andersen et al. ¹¹⁵ ; IMPACT World+; IPCC; International Reference Life Cycle Data System 2016; ReCiPe 2016; TRACI 2.0; TRACI 2.1; not stated
Hybrid (environmental) life cycle assessment (LCA)	4	Emergency medicine; gynecology and obstetrics; ophthalmology; otolaryngology	ISO 14040; ISO 14044; GHG protocol	Hybrid	TRACI 2.1; not stated	CML-IA baseline; cumulative energy demand; ReCiPe 2016; TRACI 2.1
Eye efficiency tool	4	Ophthalmology	ISO 14040; not stated	Hybrid	Eye efficiency; not stated	Not stated
Process life cycle assessment	1	Gynecology and obstetrics	ISO 14040; ISO 14044	Process	Not stated	TRACI 2.0
Process LCA and environmentally extended input-output LCA (EEIO LCA)	1	Radiology	Not stated	Hybrid	SimaPro	TRACI 2.1
Augmented process-based hybrid LCA	1	Emergency medicine	NHS' LCA-based guideline	Hybrid	SimaPro	IPCC
Hybrid life cycle analysis (LCA)	1	Ophthalmology	Not stated	Hybrid	Not stated	Not stated
Life cycle assessment (simplified)	1	Outpatient consultations in multiple medical specialties	Not stated	Process	Microsoft Excel	Not stated
Environmentally Extended Input-Output (EEIO) LCA	1	Cardiology	GHG protocol product standard	Financial	Not stated	IPCC
Economic input-output life cycle assessment (EIO-LCA)	1	Ophthalmology	Bilan Carbone protocol; GHG protocol; PAS 2050	Hybrid	Not stated	Not stated
Environment Input-Output life-cycle assessment (EIOLCA)	1	Surgery	GHG protocol	Hybrid	Not stated	Not stated
Life cycle assessment in combination with other modeling technique						
Life cycle assessment in combination with Markov model	1	Psychiatry	ISO 14040; ISO 14044	Process	SimaPro	ReCiPe 2016
Care Pathway Guidance methodology As described by Care Pathway Guidance document	1	Orthopedics	SCPG	Not stated	Not stated	IPCC
Sustainable Care Pathway Guidance tool/online calculator	2	Cardiology; Surgery	SCPG	Hybrid	SCPG	Not stated
Environmental impact assessment	1	Surgery	SCPG	Process	Not stated	Not stated
Carbon footprint method						
Bilan Carbone method	4	Gastroenterology; nephrology; ophthalmology; surgery	Bilan Carbone protocol; GHG protocol; ISO 14064-1; ISO 14067; ISO 14069; PAS 2050; not stated	Hybrid	Bilan Carbone tool	IPCC; not stated
Carbon audit	1	Gynecology and obstetrics	GHG protocol; PAS 2050	Process	Not stated	Not stated
Carbon footprint accounting/analysis/evaluation/CO ₂ production evaluation	7	Dermatology; gastroenterology; radiotherapy; surgery	GHG protocol; IPCC; ISO 14064:2018; not stated	Process; hybrid	US EPA calculator; not stated	Not stated
Greenhouse gas inventory	7	Surgery	GHG protocol	Process	Not stated	GWP100 work of Sulbaek Andersen et al. ¹¹⁶
Component analysis	4	Nephrology; ophthalmology	PAS 2050	Process; Hybrid	Not stated	Not stated

Table 3. (Continued.)

Method as described by study	Studies (n)	Medical specialty	Protocol(s)	Type(s) of activity data	Software	Characterization method(s)
Eco-audit method by Ashby ¹⁷	2	Cardiology	Not stated	Process	Ansys Granta EduPack software	IPCC; Not stated
Environmental aspect analysis	1	Nephrology	Not stated	Process	Not stated	Not stated
Multicomponent analysis	1	Surgery	Not stated	Process	Not stated	Not stated
PAS2050 methodology	1	Pathology	PAS 2050	Process	Not stated	Not stated
Top-down model of carbon emissions	1	Gastroenterology; surgery	Not stated	Financial	Not stated	Not stated
Material flow analysis						
Material flow analysis (MFA)	1	Emergency medicine	Not stated	Process	Not stated	ReCiPe 2016
Not specified						
Not specified	13	Cardiology; nephrology; ophthalmology; pediatric medicine; pulmonary medicine; radiotherapy; outpatient consultations in multiple medical specialties	GHG protocol; ISO 14064-1; PAS 2050; SCPG; not stated	Process; hybrid; not stated	R; not stated	IPCC; Not stated

Note: CO₂, carbon dioxide; EEIO-LCA, environmentally extended input-output life cycle assessment; EIO-LCA, environment input-output life cycle assessment; EPA, Environmental Protection Agency; GHG, greenhouse gas; GWP100, Global Warming Potential 100; IPCC, Intergovernmental Panel on Climate Change; ISO, International Organization for Standardization standards 14040 and 14044; LCA, life cycle assessment; NHS, National Health Service; PAS, publicly available specification; SCPG, Sustainable Care Pathway Guidance; TRACI, Tool for Reduction and Assessment of Chemicals and Other Environmental Impacts.

($n = 18$) scoring <50% (Table S22). In terms of transparency, more than half of the included studies did not clearly state the study goal and intended audience ($n = 49$), protocols followed ($n = 28$), or software used ($n = 48$). Although most studies scored high on completeness by covering multiple life cycle stages in their analyses ($n = 62$), it was not always transparently reported what elements of production were included. Many studies took emission factors for raw materials only and presented these as cradle-to-gate emissions. Regarding validity, studies rarely reported on the representativeness of the used data and the potential significance of exclusions or assumptions. Only two studies contextualized results in both sensitivity and uncertainty analyses. Limitations or the potential impact of omissions or assumptions on the study's outcomes were not adequately described by two-thirds of the studies ($n = 51$), creating a risk for bias. In half of the studies, the source(s) of funding and potential conflict(s) of interest were disclosed and unlikely to be a source of bias ($n = 38$). In terms of consistency, nearly two-thirds of the studies clearly defined and justified the functional unit consistently with the study's intended application ($n = 46$), and 82% of the studies drew conclusions consistent with both the goal and scope of the study and the impact assessment results ($n = 62$). Most studies performed contribution analyses ($n = 61$), identified hotspots ($n = 63$), clearly presented their results ($n = 65$), and had their assessments critically appraised as part of the peer-review process of the scientific journals in which the studies were published ($n = 64$).

Discussion

This review shows that the carbon footprint of an increasing number of hospital services and care pathways has been evaluated as currently reflected in the English and Dutch literature published between 2000 and January 2024. However, important medical specialties remain understudied, including internal medicine, otorhinolaryngology, and neurology. Reasons for selecting the studied topics were largely based upon volume or because no previous carbon footprint studies had been done. Methods used for carbon footprint quantification varied, and studies made different methodological choices regarding goal and scope, inventory analysis, and impact assessment. Terminology used to report on the methods was inconsistent. Factors that contributed most to the carbon footprint varied per service, pathway, medical specialty, and setting, but generally travel, facilities, and consumables were key contributors.

Implications and Suggestions for Future Research

Evaluating the climate impact of individual hospital services and care pathways provides granular information on carbon hotspots within health care and thereby enables different stakeholders, including policymakers and clinicians, to act and implement carbon reduction strategies. Although an increasing number of care topics are being studied, the task of conducting full resource-intensive LCAs for all yet unstudied and understudied services and care pathways is tremendous—especially given that results are highly context dependent. Given the urgency to rapidly move toward carbon-neutral health care systems, it is important to critically evaluate which hospital services and care pathways are most relevant for future footprint studies and how to prioritize them, for example, based on the feasibility of implementing changes to lower the carbon footprint. Presently, this prioritization is lacking, and further research should determine what types of care are deemed most relevant in this context. Furthermore, we need to consider the contexts in which the studies are performed. Currently, less than a quarter of all included studies compared similar services across different countries, hospitals, or

Table 4. Contributing factors and relative contributions ranked for each hospital service and care pathway (by row) from highest percentage (1) to lowest percentage (7), by medical specialty and hospital service or care pathway.

Medical specialty	Hospital service or care pathway	Specifics under study	Patient travel (%)	Staff travel (%)	Facilities (%)	Medical equipment (%)	Medical consumables (%)	Pharmaceuticals (%)	Waste disposal (%)	Other (%)	(ranking)	Reference(s)
Anesthesiology	Anesthesia for total knee replacement	General	—	—	—	27	28	48	(1)	28	(2) ^a	McClain et al. ²⁸
		General + spinal	—	—	—	46	27	30	(2)	27	(3) ^a	
		Spinal	—	—	—	53	23	26	(3)	23	(3) ^a	
	Anesthesia maintenance for 1 h	Desflurane	—	—	—	0.3	0.3	100	(1)	0	(2)	Sherman et al. ²⁹
Cardiology and cardiac surgery		Isoflurane	—	—	—	0 ^b	0 ^b	99	(1)	1	(2)	
		Propofol	—	—	—	0 ^b	0 ^b	81	(1)	19	(2)	
		Sevoflurane	—	—	—	0 ^b	0 ^b	100	(1)	0	(2)	
	Atrial fibrillation catheter ablation procedure	—	—	—	—	1	73	—	(3)	—	(2)	Ditac et al. ³⁰
Dermatology	Cardiac surgery	—	—	—	3	—	87	11	(1)	—	(2)	Grinberg et al. ³¹
	Cardioversion care pathway	DOAC	16	(3)	29	(2)	—	—	—	54	(1)	Orton and
		Warfarin	15	(3)	20	(2)	—	—	—	65	(1)	Pierce ³²
	Care pathway for acute decompensated heart failure	—	—	—	88	(1)	—	2	(3)	1	(4)	Zhang et al. ²²
Emergency medicine	Elective CABG with cardiac bypass	—	—	—	—	5	80	6	(1)	8	(2)	Hubert et al. ³³
	Melanoma follow-up pathway, various melanoma stages	Following 2015 guideline	4	(2)	—	—	—	—	—	96	(1)	Grover et al. ³⁴
		Following 2022 guideline	4–7	(2)	—	—	—	—	—	93–96	(1)	
	Bed dry	In ICU	—	—	—	—	86	9	(1)	—	(2)	Hunfeld et al. ³⁵
Gastroenterology		High-intensity ward	—	6	69	(1)	16	4	(4)	2	(6)	Penny et al. ³⁶
		Low-intensity ward	—	10	40	(3)	40	5	(4)	1	(5)	
		In ICU	—	6–7	21–28	(4)	27–31	2	(6)	3	(5)	Prasad et al. ³⁷
	Emergency department visit	In acute care unit	—	7–8	34–42	(5)	16–19	3	(7)	5	(6)	Penny et al. ³⁸
Gynecology and obstetrics	Treatment of septic shock in ICU	American hospital	—	6	70	(1)	6	15	(2)	0.3	(6)	McClain et al. ^{39c}
		Australian hospital	—	—	87	(1)	11	—	(4)	11	(2) ^a	
	Colonoscopy	—	—	—	76	(1)	20	—	(3)	20	(2) ^a	Elli et al. ⁴⁰
	Esophagogastroduodenoscopy/upper endoscopy	—	—	—	29	(2)	52	—	(1)	—	(3)	Elli et al. ⁴⁰
Nephrology		Without EndoFaster to reduce rate of biopsy	—	—	18	(2)	65	—	(1)	—	(2)	Zullo et al. ⁴¹
		With EndoFaster to reduce rate of biopsy	—	—	—	3	26	—	(2)	0.1	(4)	
	Gastrointestinal endoscopy	—	—	—	—	—	—	—	(3)	—	(1)	
	Medical treatment of gastroesophageal reflux	biopsy	45	(1) ^f	12	(3)	3	1	(5)	3	(6)	Lacroute et al. ⁴²
Gynecology and obstetrics	Cesarean section	—	—	—	—	—	—	40	(3)	—	(4)	Gatenby ¹⁸
	Endometrial cancer staging	Laparoscopy	—	—	58	(1)	12	—	(3)	4	(4)	Campion et al. ⁴³
		Laparotomy	—	—	62	(1) ^g	—	—	(1) ^g	40	(2)	Woods et al. ⁴⁴
	Robot-assisted Abdominal Laparoscopic	—	—	—	63	(1) ^g	—	—	(1) ^g	39	(2)	
Nephrology	Hysterectomy	—	—	—	65	(1) ^g	16	64	(2)	0	(3)	Thiel et al. ²¹
		Robotic	—	—	11	(3)	59	29	(1)	0	(4)	
		Vaginal	—	—	6	(3)	64	28	(1)	0	(5)	
	Vaginal birth	—	—	—	4	(3)	14	69	(2)	2	(5)	Campion et al. ⁴⁵
Nephrology	Anbulatory peritoneal dialysis	Continuous, at home center	3	(5)	58	(1)	5	—	(4)	6	(3)	Chen et al. ⁴⁵
		Continuous, in PD center	0	—	7	(3) ^g	79	—	(1)	8	(2)	
		Daytime, at home center	2	(5)	9	(2) ^g	89	—	(1)	—2	(4)	
		Daytime, in PD center	0	—	7	(3) ^g	79	—	(1)	4	(3)	

Table 4. (Continued.)

Medical specialty	Hospital service or care pathway	Specifics under study	Patient travel (%)	Staff travel (%)	Facilities (%)	Medical equipment (%)	Medical consumables (%)	Pharmaceuticals (%)	Waste disposal (%)	Other (%)	(ranking)	Reference(s)
Ophthalmology	Bed day nephrology ward	With Prismaflex system	2	10	8	—	—	—	58	71	(1)	Connor et al. ⁴⁶
	Continuous renal replacement therapy (per 72 h)	At home	—	—	—	20	78	—	—	2	(3)	Aspin ⁴⁷
	Hemodialysis (per year)	In center	0–1	0–2	2–60	2–60	37–60	—	3–16	2–8	(4)	Connor et al. ⁹⁸
		At home	20	5	31	31	37	—	3	5	(4)	James ⁴⁸
		In center	—	—	100	—	—	—	—	—	—	
		At home	68	—	32	—	—	—	—	—	—	
		In hospital	6	3	26	23	23	36	3	3	(7)	Lim et al. ⁵⁰
		In center	13	9	45	45	—	—	6	27	(2)	Mitoui et al. ⁴⁹
		In center	27	5	32	2	10	18	6	—	—	Newcastle upon Tyne Hospitals ⁵¹
		In center	31	31	43	43	9	—	17	—	—	Seghal et al. ²⁰
Ophthalmology	Hemodialysis (per treatment)	—	—	—	—	—	—	—	—	—	—	
	Hemodialysis and peritoneal dialysis	26	26	14	—	—	—	—	13	47	(1)	Connor et al. ⁴⁶
	Outpatient appointment	46	16	9	4	—	—	—	3	27	(2)	
	Cataract surgery	9	0.08	1	3	3	73	13	1	—	—	Ferrero et al. ⁵²
	Phacoemulsification	27–72	27–72	0–53	1–9	4	6–67	—	0–3	—	—	Goel et al. ⁵³
	Phacoemulsification	26	26	21	6	4	45	—	2	—	—	Hong et al. ⁵⁴
	Phacoemulsification	8	12	1	68	1 ^b	68	10	0.1	—	—	Latta et al. ⁵⁵
	Phacoemulsification at public hospital	7	4	1	85	1 ^b	85	4	0.1	—	—	
	Phacoemulsification in private hospital	11	11	4	—	—	78	7	0.4	—	—	Pascual-Prieto et al. ⁵⁶
	Phacoemulsification	—	—	22	63	1	15	4	—3	0	—	Thiel et al. ⁵⁷
Orthopedics	Manual small incision	27–71	27–71	1–54	3	1–9	3–65	—	0–4	—	—	Goel et al. ⁵³
	Cataract pathway	7	3	36	33	2 ^b	33	18	2	1	(7)	Morris et al. ¹⁹
	Intravitreal injection	40	40	14	—	—	45	45	1	—	—	Chandra et al. ⁵⁸
	Total knee replacement care pathway	77	—	4	—	—	19	—	0.04	—	—	Power et al. ⁵⁹
	With Care4Today program	3	—	—	—	—	—	—	—	97	(1)	Johnson & Johnson ⁶⁰
	Without Care4Today program	3	—	—	—	—	—	—	—	97	(1)	
	Total knee replacement surgery	17	10	0.2	29	1	28	—	15	—	—	Delate et al. ⁶¹
	Tonsillectomy	—	—	1	1	3 ^g	31	67	31	1	(3)	Meiklejohn et al. ⁶²
	Coblation	—	—	—	1	3 ^g	10	88	10	1	(3)	
	Cold excision without cautery	—	—	—	1	3 ^g	10	89	10	1	(3)	
Pediatric medicine	Monopolar electrocautery	—	—	—	1	3 ^g	10	89	10	1	(3)	
	With SmartInhaler	7	—	—	—	—	—	72	—	20	(2)	Budgen ⁶³
	Without SmartInhaler	2	—	—	—	—	—	53	—	45	(2)	
	ABG	—	—	—	30	2	70	—	70	—	—	McAlister et al. ⁶⁴
	CRP	—	—	—	23	2	78	—	78	—	—	
	Coagulation profile	—	—	—	63	1	37	—	37	—	—	
	Full blood examination	—	—	—	22	2	78	—	78	—	—	
	Urea + electrolytes	—	—	—	58	1	41	—	41	—	—	
	Phlebotomy	—	—	—	23	2	76	—	2	—	—	Spoyalo et al. ⁶⁵
	Chemistry	—	—	—	79	1	20	—	1	—	—	
Pathology	Coagulation factors	—	—	—	3	3	92	—	4	—	—	
	Hematology	—	—	—	4	3	90	—	6	—	—	
	Total protein	—	—	—	53	1	45	—	2	—	—	
	Urinalysis	—	—	—	35	2	65	—	65	—	—	McAlister et al. ⁶⁴
	Urine testing	—	—	—	—	—	—	—	—	—	—	

Table 4. (Continued.)

Medical specialty	Hospital service or care pathway	Specifics under study	Patient travel (%)	Staff travel (%)	Facilities (%)	Medical equipment (%)	Medical consumables (%)	Pharmaceuticals (%)	Waste disposal (%)	Other (%)	Ranking (ranking)	Reference(s)
Psychiatry	COVID-19 testing	One-monthly	—	—	—	—	28	(2)	—	—	(1)	Ji et al. ⁶⁶
	Schizophrenia pathway with antipsychotic injections	Three-monthly Treatment interruption	53 (1)	0.4 (3)	44 (4)	—	—	—	—	—	(3)	Debayve et al. ⁶⁷
			67 (2)	0.6 (3)	69 (3)	—	—	—	—	—	(3)	
Pulmonary medicine	Bronchoscopy	—	—	—	—	23	—	—	77	—	(1)	Patrucco et al. ⁶⁸
	Monoclonal antibody therapy of severe eosinophilic asthma	With Benralizumab Without Benralizumab	4 (3)	—	—	—	—	—	—	35	(2)	Budgen ⁶⁹
	CT scan	—	—	—	—	—	—	62	—	—	(2)	
Radiology	Chest X-ray	Mobile	—	—	17	83	9	—	—	—	(2) ^a	Martin et al. ⁷⁰
			—	—	—	88	15	—	15	4	(3)	McAlister et al. ⁷¹
			—	—	—	3	9	—	9	—	(2) ^a	
Radiotherapy	Interventional radiology procedure	—	—	2	50	4	41	0.08	2	88	(1)	Chua et al. ⁷²
	MRI scan	—	—	—	2	95	1	—	9	1	(5)	Esmaili et al. ⁷³
	Ultrasound	—	—	—	13	88	1	—	—	3	(2)	Martin et al. ⁷⁰
Radiotherapy	External beam radiation therapy	—	—	—	19	94	2	—	2	5	(2)	McAlister et al. ⁷¹
			—	—	—	80	12	—	12	—	(2) ^a	Martin et al. ⁷⁰
			—	—	—	88	3	0	2	10	(4)	Ali and Piffoux ⁷⁵
Surgery	Proton therapy	—	—	—	—	0	1	—	—	—	—	Cheung et al. ⁷⁶
	Abdominoplasty	During pandemic	14 (3)	18	53	83–84	0.1	—	—	—	—	Chuter et al. ⁷⁷
	Anastomotic leak care pathway	Pre-pandemic Prostate VMAT, pre-COVID	0.5 (3)	—	—	9	23	—	—	—	—	
Surgery	Anti-reflux surgery	American hospital	—	—	—	—	—	3	—	—	—	Dvorak et al. ⁷⁸
	Average surgery	Canadian hospital	—	—	36	1	12	51	12	1	(3)	Berner et al. ⁷⁹
		UK hospital	—	—	16	0	19	64	19	2	(4)	Bischofberger et al. ⁸⁰
Surgery	Bilateral breast augmentation	In LMIC	—	—	83	3	11	4	11	0	—	Gatenby ¹⁸
	Carpal tunnel release	Standard model	21 (2)	17	49	8	—	31	0.2	2	(6)	MacNeill et al. ⁸¹
		Lean and green	—	—	21	—	—	44	35	—	—	Penny et al. ⁸²
Urology	Elective endovascular aortic repair	Open	—	—	—	65	35	—	3	2	(5)	Umo et al. ⁸³
	Minimally invasive surgery	Hospital-based	—	—	—	—	10	—	—	—	—	Berner et al. ⁷⁹
	Rhinoplasty	Clinic-based	20 (2)	16	49	11	—	—	—	—	—	Kodumuri et al. ⁸⁴
Urology	Skin cancer surgery	Prebiopsy prostate MRI	75 (1) ^f	75 (1) ^f	17	7	3	—	2	2	(5)	Zhang et al. ⁸⁵
	Prostate biopsy pathway	TRUS prostate biopsy	28 (2) ^f	28 (2) ^f	65	0	7	—	—	0	—	Senéaud et al. ⁸⁶
		Pathology analysis	22 (2) ^f	22 (2) ^f	65	65	11	—	0	0	—	Power et al. ⁸⁷
Urology	Radical prostatectomy	Laparoscopic	—	—	—	56	11	—	1	0	—	Berner et al. ⁷⁹
			—	—	5	5	64	—	16	0	—	Tan and Lim ⁸⁸
			—	—	78	78	22	—	22	—	(2) ^a	Leapman et al. ⁷⁴

Table 4. (Continued.)

Medical specialty	Hospital service or care pathway	Specifics under study	Patient travel (%)	Staff travel (%)	Facilities (%)	Medical equipment (%)	Medical consumables (%)	Pharmaceuticals (%)	Waste disposal (%)	Other (%)	(ranking)	Reference(s)
Outpatient consultations in multiple medical specialties	Face-to-face consultation	Robot-assisted Geriatric Medicine clinic	60	(1)	35	(2)	80	(1) ⁸	20	(2) ⁸	—	Bartlett and Keir ⁹⁰
		Urology department	>99	(1)	—	—	<1	(2)	—	—	1	Filfilan et al. ⁹¹
		Rehabilitation clinic	100	(1)	—	—	—	—	—	—	—	Holmer et al., 2014 ⁹²
		Speech therapy clinic	100	(1)	—	—	—	—	—	—	—	—
	Neuroemergent consultation	Psychiatry clinic	100	(1)	—	—	—	—	—	—	—	Penaskovic et al. ⁹³
		Benign foregut clinic	99.5	(1)	—	—	0.1	(3)	0.5	(2)	—	Sillcox et al. ⁹⁴
		Without telehealth	99	(1)	—	—	—	—	<1	(3)	—	Thiel et al. ⁹⁵
		With telehealth	50	(1)	50	(1)	—	—	—	—	—	Whetten et al. ⁹⁶
	Preoperative screening	Before telehealth	44	(2)	—	—	0.02	(2)	—	—	—	Wang et al. ⁹⁷
		After telehealth	51	(1)	—	—	—	—	—	—	—	—
Virtual consultation	Geriatric medicine clinic	Implementation	—	—	—	—	—	—	—	—	—	—
		Implementation	—	56	(1)	(3) ⁸	7	(3) ⁸	0	—	—	Bartlett and Keir ⁹⁰
		Urology department	—	—	—	—	100	(1)	—	—	—	Filfilan et al. ⁹¹
		Rehabilitation clinic	—	—	—	—	100	(1)	—	—	—	Holmer et al. ⁹²
	Speech therapy clinic	Speech therapy	—	—	—	—	100	(1)	—	—	—	—
		Psychiatry clinic	—	—	—	—	100	(1)	—	—	—	Penaskovic et al. ⁹³
		Benign foregut clinic	—	—	—	—	100	(1)	—	—	—	Sillcox et al. ⁹⁴
		By phone	—	—	—	—	100	(1)	—	—	—	Thiel et al. ⁹⁵
	By video	By video	—	—	—	—	100	(1)	—	—	—	—
		—	—	—	—	—	—	—	—	—	—	—

Note: Results should not be directly compared between studies and services owing to differing methodologies. Data on relative contributions were obtained from original studies. Results were summarized and reported in the following categories: patient travel; staff travel; facilities (e.g., heating, lighting, water use, construction and maintenance of building); medical equipment (e.g., MRI scanner, hemodialysis machine); medical consumables (e.g., disposable instruments, syringes, dressings, gloves, masks); pharmaceuticals (e.g., anesthetic gases, medical drugs, pharmaceutical packaging); waste disposal (e.g., waste incineration, landfill, recycling, wastewater treatment); other (e.g., nonmedical consumables, food, office supplies; nonmedical equipment, computers). For some services, it was not possible to group a factor to one category, because only aggregate percentages were presented in the original study. In those cases, the total contribution of this was attributed to each applicable category (i.e., double counting), meaning that row totals could be >100%. —, Not applicable; ABG, arterial blood gas; CABG, coronary artery bypass graft; COVID, coronavirus disease; CRP, C-reactive protein; CT, computerized tomography; DOAC, direct oral anticoagulant; ICU, intensive care unit; IMRT, intensity-modulated radiotherapy; LMIC, low- or middle-income country; MRI, magnetic resonance imaging; PD, peritoneal dialysis; TRUS, transrectal ultrasound; VMAT, volumetric modulated arc therapy.

⁸This percentage included both medical consumables and waste disposal.

⁹This percentage included both medical equipment and medical consumables.

¹⁰This percentage included both staff and visitor travel.

¹¹This percentage also included nonmedical consumables (food).

¹²These studies quantified the carbon footprint of pharmaceuticals, but only presented results in separate analyses.

¹³This percentage included both patient and staff travel.

¹⁴This percentage both included facilities and medical equipment.

¹⁵This percentage included both medical equipment and medical consumables.

¹⁶This percentage included patient, staff, and visitor travel.

¹⁷This percentage also included both consumables and pharmaceuticals (non-injected medications).

¹⁸This percentage included both patient, staff, and visitor travel and transportation of consumables.

¹⁹This study also included variations to the care pathway, with different scenarios and prostate biopsy sampling approaches (not presented in this table).

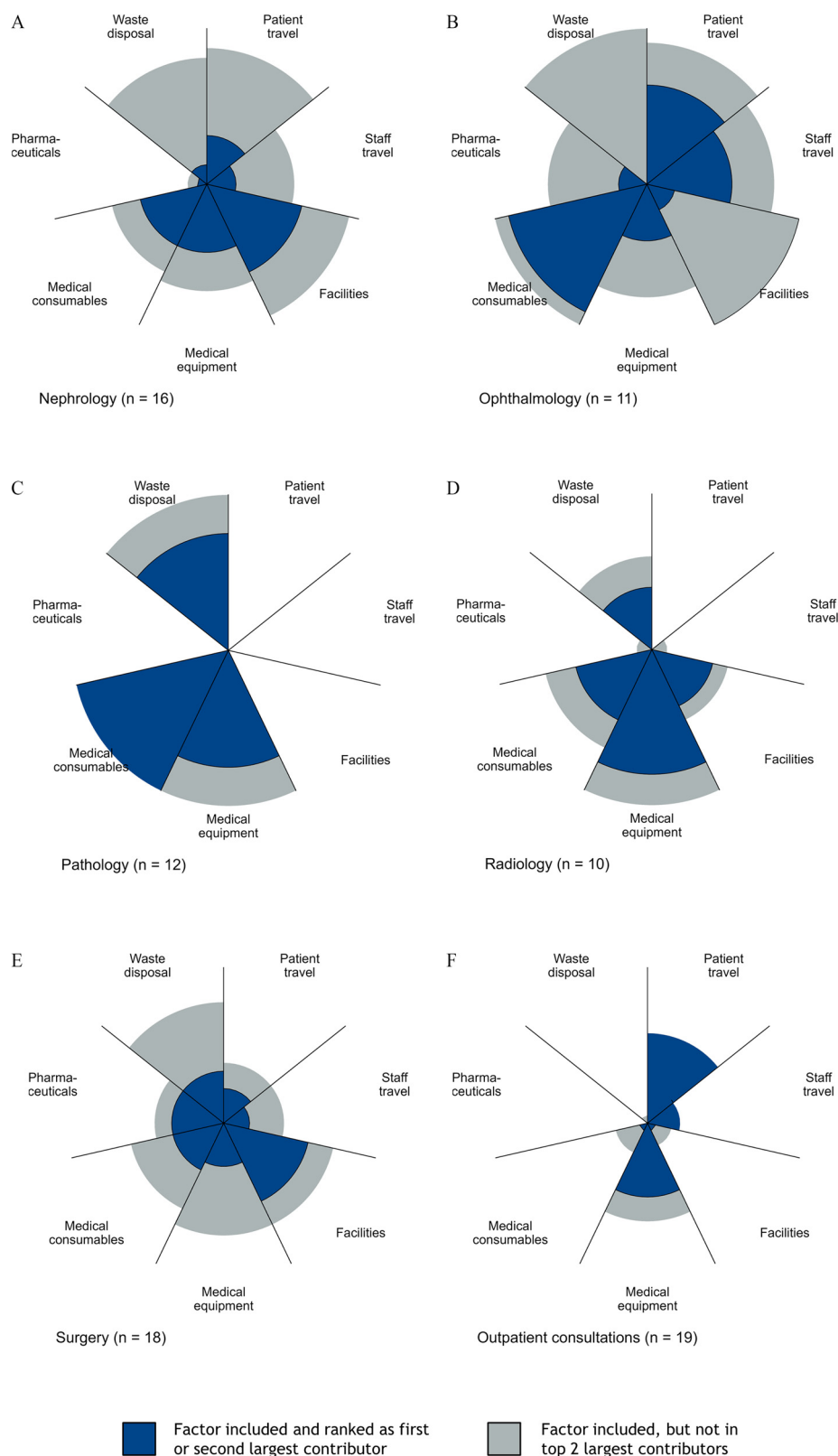


Figure 2. (A–F) Share of hospital services and care pathways that included specific contributors of the carbon footprint, presented for six different medical specialties. Medical specialties were graphically depicted if >9 individual services or pathways were studied among all sources identified. Data used for these graphs can be found in [Table 4](#). The end of each black line indicates 100% of services/pathways represented in the figure. This figure should be interpreted with caution, given that it reflects only services/pathways that have currently been studied within these specialties and is likely not fully representative of these medical specialties as a whole. An empty area does not indicate that contributors had zero impact but only shows that it was not included for evaluation in the services/pathways studied within these specialties (i.e., missing data). Given that the chart represents individual services/pathways and not studies themselves, this figure also risks overvaluing those studies that included multiple services/pathways.

settings. These studies generated important insights about the variation in carbon footprints and hotspots, revealing differences in medical practice around anesthetic use and energy efficiency at different locations. Comparative studies provide a great opportunity for understanding what explains the variation, for identifying best practices to lower climate impact, and may also provide insights for potential transferability of results.

Although individual carbon hotspots differed per hospital service, care pathway, medical specialty, and setting, some common factors contributed to the footprint of many services and pathways, including travel, the facilities' and medical equipment's energy use, and medical consumables. Other research on national-level health systems shows that Scope 3 upstream supply chain emissions from pharmaceuticals, medical equipment, and private travel generally contribute most to the carbon footprint of hospital care, followed by Scopes 1 and 2 building emissions from energy use.^{4,9,99} Our study results revealed similar hotspots, although the relative size and importance of their contributions varied from national estimations because of differences in methodologies, study scopes, and data availability. This suggests a dual-track approach is possible: We should target mitigation strategies based on both *a*) common patterns (e.g., greener forms of energy because of their large contribution to the carbon footprint of many hospital services), and *b*) specific hotspots within a certain service, pathway, specialty, or setting.^{30,45} For example, stakeholders involved in organizing or providing services that require much patient travel, such as outpatient consultations, may prioritize initiatives that aim to reduce travel-related emissions. However, medical specialties that mostly provide services with a high contribution of facilities' energy use, such as surgery and nephrology, may better focus on reducing or optimizing energy use per service. Importantly, in both examples, the setting should also direct initiatives, because priorities may lie elsewhere in a modernized, well-isolated hospital building or in outpatient departments situated near public transportation.

Evidently, a hotspot can be found only if it was included in the first place. Pharmaceuticals were the least included category among the studies in this review, despite it being one of the major contributors to health care systems' national footprints.^{4,10} This may be explained by a lack of life cycle inventory data for pharmaceuticals.¹⁰⁰ Not including potentially relevant factors may leave important contributors unseen or distort the analysis, making the relative share of certain contributors appear larger than it actually is. This in turn may direct carbon reduction efforts toward less impactful areas. On the other hand, including all products and processes may be very time consuming, and standards state that it can be justified to exclude an input if it is environmentally insignificant in the context of the study's goal.^{101,102} Although it can be difficult to determine beforehand what inputs will not significantly change results, the studies included in our review showed which products and processes generally had marginal contributions to the carbon footprint of multiple hospital services and care pathways, such as paper and office supplies. Although ultimately also goal and context dependent, future footprint studies of hospital services and care pathways might better focus time and efforts on data collection for factors with generally larger contributions (e.g., facilities energy use, travel) or understudied but likely important contributions (e.g., pharmaceuticals) rather than counting every paper towel individually.

Our results show that carbon footprint studies within health care apply different methods and make different methodological choices (e.g., about functional units and system boundaries). In the majority of studies, the functional unit did not include the entire care pathway. In nonmedical sectors, extensive product rules and sector guidance documents exist to allow for a fair comparison

across products. It is important to create consensus regarding used methodologies within health care as well, which has been argued by others before.¹³ A first step toward methodological consensus is to develop a common vocabulary in health care LCA research. The currently prevailing use of different terminologies and abbreviations may create confusion. Furthermore, future studies should be more transparent regarding data sharing, especially life cycle inventory data, and their reporting on the influence of modeling choices on the study's outcomes and conclusions. Although it is imperative not to delay the reduction of the carbon footprint associated with hospital services and care pathways, establishing consensus and enhancing methodological quality is crucial. This is particularly important if we intend to use the climate impact of hospital services as a factor in selecting between clinical alternatives during medical decision-making.

Strengths and Limitations

In this study, we focused on hospital care because this is one of the most energy- and carbon-intensive forms of health care delivery.⁹⁹ We conducted a complete state-of-the-science review of all carbon footprint studies of hospital services and care pathways, including gray literature. By including care from multiple medical specialties and extending the focus from individual hospital services to entire care pathways, we provided a more comprehensive overview of evidence compared with other studies that focused on a specific selection of hospital care, such as surgical and anesthetic care,^{14,15,103,104} gynecology,¹⁰⁵ dermatology,¹⁰⁶ orthopedics,^{107–109} and radiology,¹¹⁰ or studies that did not include gray literature.¹¹¹ Furthermore, to our knowledge, this state-of-the-science review is the first study to assess the quality of many of these studies.^{13,111}

A limitation of the current field of health care sustainability research is that meta-analyses cannot be undertaken, which is a common issue for LCA reviews.¹² Methods are heterogeneous, and results are inconsistently presented on, for example, different levels of aggregation. Although our review provided an overview of several methodological choices in LCA—including goal and scope definition, inventory analysis, and impact assessment—some important methodological choices remain to be researched. The identified footprints and hotspots each have numerous underlying data sources, assumptions, and allocation procedures. Further research is needed to develop a better understanding of these methodological choices and work toward methodological standardization.

A limitation of our study could be the relatively narrow focus on carbon footprint, which is only one environmental impact category. Basing sustainability decisions purely on one environmental impact risks aggravating other impacts, such as resource use or marine ecotoxicity. Despite the impact of hospital care on the environment being much broader than its carbon footprint, the majority of environmental impact studies within hospital care included only carbon footprint.¹¹² Given this lack of information on other important environmental impact factors, future studies should also address other environmental impact categories, such as those included in the ReCiPe 2016 method.¹¹³ Furthermore, although our broad study scope extended the focus from individual hospital services to entire hospital care pathways, it must be mentioned that nonhospital care also carries a footprint.⁹⁹ Ideally, future research should include the whole patient trajectory, including relevant care provided outside of the hospital. Despite some limitations in the search process, where we could have included other search terms such as “climate change” or “waste,” our search was sensitive to the relevant studies and even identified studies that were not included in the comprehensive Healthcare LCA database.²³ Finally, a limitation is that the *pro forma* tool¹⁵ we used for quality assessment may not have been the most suitable for all footprint

studies, given that it had a specific focus on attributional LCA methodology, and many studies applied different approaches and followed different protocols. As far as we know, no standardized assessment tool for the reporting of carbon footprint studies in health care exists to date, and further research should work on standardized transparency catalogs synthesizing multiple guidelines.¹¹⁴

Conclusion

This state-of-the-science review shows that the carbon footprint of an increasing number of hospital services and care pathways are being studied. These services and pathways play a central role in hospital care delivery. Factors that contributed most to the carbon footprint of hospital services varied per service, medical specialty, and setting, but generally travel, facilities, medical equipment, and consumables were important contributors. The variability in carbon footprints across different settings underlines the importance of conducting local studies in various settings and tailoring sustainability efforts accordingly. Standardization of carbon footprint methodologies, terminology, and reporting is needed to further develop the field of health care sustainability research.

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